EE5175 Assignment 1

Amritam Halder ee23b004

6th February 2026

1 Geometric Transformations

This lab assignment gives a learning experience on a few basic geometric transformations in images. Learned the true way to scale and rotate an image for better noiseless feature extraction. Used **binary interpolation** for better target-to-source mapping.
The coordinate convention followed is as follows:

- Origin(0,0) is in the top-left corner of the image.
- X-axis denotes the rows pixels.
- Y-axis denotes the columns pixels.

1.1 Translation

Translation means a shift in the position of the pixels in the image by (tx, ty). The translation matrix in homogeneous coordinates is given as :

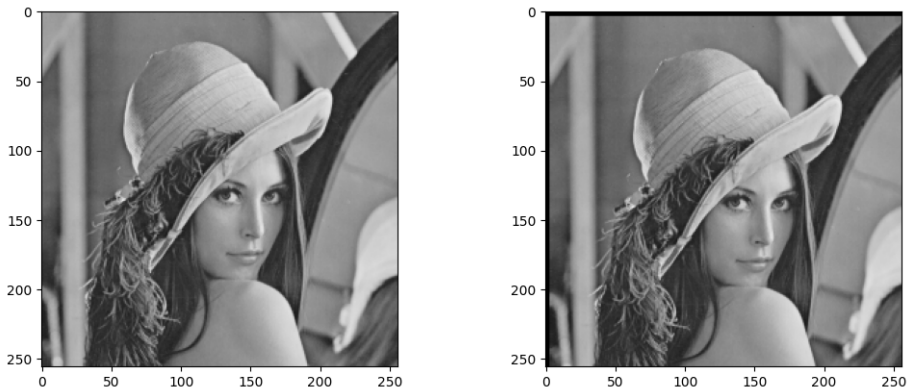
$$T = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix}$$

The source pixel in homogeneous coordinate is given by:

$$S = \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

The translation of the image from a source (x_s, y_s) to a target (x_t, y_t) is performed as

$$S = T^{-1} \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix}$$



(a) Original (b) Translated

Figure 1: Effect of translation on Lena image

Thus:

$$\begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix} = \begin{bmatrix} x_t - t_x \\ y_t - t_y \\ 1 \end{bmatrix}$$

1.2 Rotation

A rotation of the image is defined as the angular shift of the image pixel position. The rotation matrix for an angle θ is given as:

$$R = \begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Let the target pixel position be given as is given as:

$$T = \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix}$$

The Rotation is done as:

$$S = R^{-1}T$$

This is valid if the center of the image is origin, so for our coordinate convention, we first need to translate the image center (width/2, height/2) to origin(0,0). For the leaning tower of pisa image, it is rotated by 3.8 degrees to straighten it.

(a) Original

(b) Translated

Figure 2: Effect of Rotation on Pisa Tower image

1.3 Scaling

Scaling of image is a useful way for better feature visibility. The scaling matrix of the image is given as:

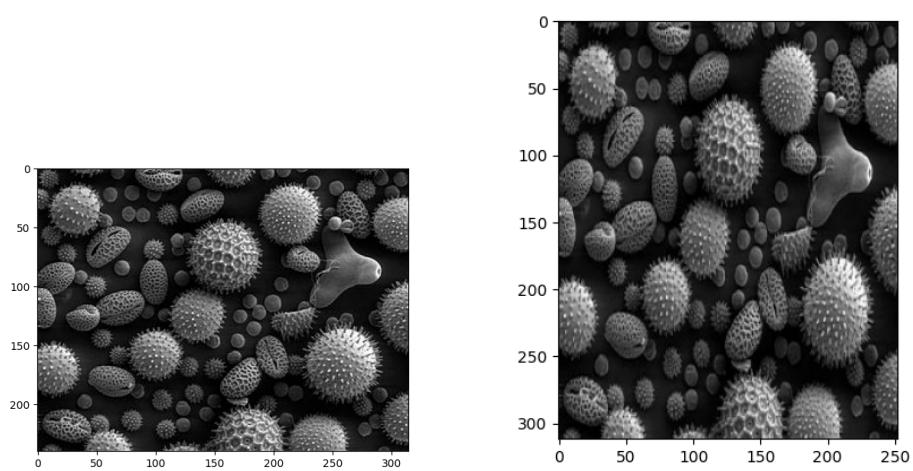
$$\alpha = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now for scaling of an image, few pixels position may get skipped without having a definite Intensity value. The issue can be solved by doing target-to-source mapping and then using binary interpolation to find the intensity for every pixel in the scaled image. The current implementation of binary interpolation is as follows:

$$x_s = \frac{x_t}{s_x}, y_s = \frac{y_t}{s_y} \tag{1}$$

$$a = x_s - \lfloor x_s \rfloor, b = y_s - \lfloor y_s \rfloor \tag{2}$$

$$\begin{aligned} I_t(x_t, y_t) = &(1-a)(1-b)\, I_s(\lfloor x_s \rfloor, \lfloor y_s \rfloor) + (1-a)b\, I_s(\lfloor x_s \rfloor, \lfloor y_s \rfloor + 1) \\ &+ a(1-b)\, I_s(\lfloor x_s \rfloor + 1, \lfloor y_s \rfloor) + ab\, I_s(\lfloor x_s \rfloor + 1, \lfloor y_s \rfloor + 1) \end{aligned} \tag{3}$$



(a) Original

(b) Translated

Figure 3: Effect of Scaling on Cell image by 0.8 on rows and 1.3 on columns